

Department of Mechanical and Production Engineering

Aarhus University

Assignment 2: Mechanical Vibrations

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Lecture 1: Single-Degree-Of-Freedom Vibration

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Problem 1

A mass m is connected to a linear spring with stiffness k . Neglect damping.

- (a) Derive the governing equation.
 - (b) Determine the natural frequency.
-

Assumptions

- The spring is linear.
- Motion is one-dimensional.

Givens

$$\begin{array}{l|l} m & 2 \text{ kg} \\ k & 18 \text{ N m}^{-1} \end{array}$$

Derivation

- (a) Newton's second law gives

$$m\ddot{x} + kx = 0.$$

Result

$$m\ddot{x} + kx = 0$$

- (b) Comparing with the standard oscillator equation gives

$$\omega_n^2 = \frac{k}{m}.$$

Result

$$\omega_n = \sqrt{\frac{k}{m}} = 3 \text{ rad s}^{-1}$$

Final result – Problem 1**(a)**

$$m\ddot{x} + kx = 0$$

(b)

$$\omega_n = \sqrt{\frac{k}{m}} = 3 \text{ rad s}^{-1}$$